Taylor J Turner

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Software and electrical controls engineer with 10+ years of experience in industrial automation, spanning robotics software, machine vision, and electrical controls. Delivers integrated systems from prototype through production deployment across customer sites.

Work Experience

Toyota Advanced Logistics *(formerly Bastian Solutions: Robotics)*
Software Engineer II *April 2025 – Present*
Software Engineer I *September 2022 – April 2025*

- Led cross-functional integration of robots, vision systems, and PLCs, enabling real-time automated logistics workflows.
- Automated research and development (R&D) model-training and initial-deployment workflows using containerization and CI/CD, reducing turnaround time by 40%.
- Delivered automation platforms from prototype through production deployment across multiple customer sites, coordinating with customers, engineers, and field technicians to commission and troubleshoot integrated systems.

Selected Projects

- **PickAndDrop** (Python, Angular): Developed middleware connecting a vision system to a FANUC robot, implementing a custom detection pipeline for pallet pocket localization and socket communication to automate pallet handling.
- **SmartPick** (Python, Angular): Integrated an in-house instance segmentation model trained in PyTorch into a robotic bin-picking vision system; built Azure annotation tools and a dataset-management interface.
- **ImagePicker** (Python, React): Connected warehouse order systems to robots using optical character recognition (OCR) and human interface device (HID) automation.
- **StackOrder** (C#, React): Modernized .NET Core pallet-pattern generation software, supporting real-time pattern visualization and editing along with multiple peripheral devices.

Beckwood Press
Electrical Controls Engineer II *October 2021 – September 2022*
Electrical Controls Engineer I *October 2016 – October 2021*

- Led electrical system design including PLC programming, UL 508A-certified panels, and customer acceptance testing.
- Standardized machine parameterization and control programs, reducing configuration errors and improving deployment consistency.

Selected Projects

- **Hydraulic Control Systems:** Achieved cylinder position control within 0.010 inches and force accuracy within ±1% using feed-forward algorithms and Kalman filtering.
- **Servo Press Platforms:** Delivered high-precision all-electric presses using Yaskawa and Allen-Bradley servo systems.
- **Auxiliary Automation Systems:** Designed integrated infeed/outfeed, containment, and conveyance solutions.

EPIC Inc.
Electrical Controls Technician *December 2014 – October 2016*

- Assembled electrical panels and industrial systems for pilot plants, vision systems, and custom equipment.
- Implemented full electrical installations, including panels and field devices, following engineering designs for NEC, sanitary/wash-down, and Class I, Division 1 requirements.
- Supported engineering teams during startup and factory acceptance testing to verify system operation.

Skills

- Languages: Python, C#, C++, TypeScript, T-SQL, Structured Text (IEC 61131-3)
- Backend: FastAPI, SQLAlchemy, ASP.NET Core, Entity Framework, gRPC
- Machine Vision / ML: PyTorch, Instance Segmentation, OCR
- Frontend: Angular, React, Tailwind CSS, HTML5, CSS3
- DevOps / Cloud: Azure, Docker, CI/CD, Vercel, Neon, Render
- Automation Platforms: Studio 5000, TwinCAT, TIA Portal, Ignition, Indusoft

Education

Ranken Technical College — Associate Degree in Electrical Control Systems December 2014

Community & Interests

Merit Badge Counselor (Radio) — Scouting America
Amateur Radio Operator — Call Sign: AFØFR, Projects: af0fr.com *Extra Class License*
Football Coach (Offensive/Defensive Line) — Titans of Troy, IL